

Rafael Amaral dos Santos Pinto

Data Analyst | Data Engineer · Generative AI · M.Sc. Candidate, CEFET/RJ

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🌐 Professional Portfolio — Projects 📄 Lattes CV



AZ-900



Oracle



English B1



Cisco



AWS

SUMMARY

IT professional with 8+ years in the market, working since 2023 as a Data Analyst at Múltipla TI. I build ETL/ELT pipelines, data modeling, and Generative AI solutions (LLMs, RAG, AI Agents) in production on Microsoft Fabric, Node.js, Python, and SQL. I combine technical autonomy with close engagement with multiple corporate clients, translating business needs into data solutions. M.Sc. candidate in Computer Science (CEFET/RJ), researching TinyML: optimizing Machine Learning models for constrained hardware, applying the same efficiency discipline to reduce cost and latency in production AI systems.

PROFESSIONAL EXPERIENCE

Múltipla Tecnologia da Informação Ltda — Rio de Janeiro, RJ

Jan/2023 – Present

Data Analyst (promoted from IT Support Technician in Sep/2025)

- Develop and maintain in production RAG (Retrieval-Augmented Generation) pipelines and AI Agent architectures on top of LLMs, with serverless REST APIs in Node.js 18/20 and Azure Functions, PostgreSQL 18 + pgvector vector database for embedding indexing, and corporate authentication via Azure AD B2C
- Develop ETL/ELT, ingestion, and orchestration processes for data pipelines in Microsoft Fabric, organizing Lakehouses into layers (medallion architecture: bronze, silver, gold) and modeling Semantic Models in star schema (fact and dimension tables) with Direct Lake, in addition to administering Capacities in a multi-tenant production environment, with governance and per-client access control, integrating REST APIs and Microsoft Graph API
- Implemented an observability and monitoring engine for AI in production with LangSmith, Sentry, and Redis/ioredis, with model call tracing, execution traceability, and failure handling
- Perform ingestion and integration of data from multiple heterogeneous sources (SQL Server, Oracle, REST APIs) into a unified analytics layer, with data modeling and optimized SQL queries, simultaneously serving multiple corporate clients in retail, services, and technology
- Ensure availability and observability of data environments with Zabbix and Grafana across segregated environments (dev/beta/staging/production), with proactive monitoring and incident handling
- Built n8n automation that replaces manual alert checking with direct database detection every 10 minutes, cross-referencing two PostgreSQL databases to identify the root cause and enrich the team notification, eliminating the manual investigation that was previously required
- Version and ship pipeline and service code with Git/GitHub and Azure DevOps, with automated testing (Jest) and controlled promotion across environments
- Automate integrations and operational routines with n8n, Webhooks, and Selenium, reducing recurring manual work in data processes
- Collaborate with the development team on the AI module of the proprietary platform, with a Git workflow based on feature branches and merges

LMG Bonfim Serviço e Apoio Administrativo Ltda — Rio de Janeiro, RJ

Dec/2017 – Jan/2023

IT Support Technician

- Built reports and operational KPIs from internal systems data, supporting management decision-making
- Integrated POS systems and remote infrastructure in a corporate environment, with multi-store technical support
- Analyzed and handled technical support tickets, prioritizing critical incidents

Administrative Assistant

- Extracted and processed ERP system data for inventory control and recurring management reports
- Supported asset maintenance and basic IT support

EDUCATION

M.Sc. in Computer Science — Research line: TinyML

CEFET/RJ — Centro Federal de Educação Tecnológica Celso Suckow da Fonseca · In progress, expected completion 2028

Postgraduate Degree in Data Science and Artificial Intelligence

UNISUAM — Centro Universitário Augusto Motta · Completed, Dec/2025

Bachelor's Degree in Systems Analysis and Development

UNISUAM — Centro Universitário Augusto Motta · Completed, Dec/2024

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals (AZ-900) — Microsoft, 2026

Oracle Data Platform Foundations Associate (1Z0-1195) — Oracle, 2025

EF SET English Certificate, Level B1 — EF SET, 2025

Introduction to Cybersecurity — Cisco Networking Academy, 2023

AWS Academy Graduate — Cloud Foundations — AWS Academy, 2022

PROJECTS

Consolidador de Consumo SKU — Microsoft Fabric Capacity Metrics

Before this system, capacity sizing for clients in POC/trial was done through manual estimation, with no standardized criteria. I built the algorithm and the underlying consumption data analysis from scratch: a tool that replicates Fabric's own smoothing and allocation algorithms to calculate real consumption per SKU (F2–F2048) and recommend the ideal SKU with precision. It now serves multiple clients simultaneously, reducing the time the sales team spends assembling proposals and increasing sizing accuracy. One of the largest systems I have built at Múltipla TI, with automatic PDF proposal generation (Puppeteer) and SharePoint integration via Microsoft Graph API

Robô Terrestre Híbrido — Simulação 3D

rafaelpinto1.github.io/cefet-robotica

Final project for the Robotics Applications course (M.Sc. program): autonomous differential-drive robot with a hybrid deliberative-reactive architecture, Arduino Uno, and interactive 3D simulation in Three.js.

TECHNICAL SKILLS

Artificial Intelligence & AI Engineering — LLMs, Generative AI, RAG, vector database, AI Agents, Prompt Engineering, Computer Vision, OCR, Embeddings, TinyML, LangSmith

Data Engineering & Analytics — Microsoft Fabric (Lakehouse, Data Pipelines, Semantic Models, Direct Lake, Capacities), Medallion Architecture (Bronze/Silver/Gold), Dimensional Modeling (Fact/Dimension, Star Schema), ETL/ELT, Data Modeling, Data Pipelines, Advanced SQL, T-SQL, Power BI, Advanced DAX, Power Query M, Python, PostgreSQL 18, pgvector, SQL Server, Oracle

Development & Systems Integration — REST APIs, Microsoft Graph API, Git/GitHub, Azure DevOps, Azure Functions, serverless architecture, Azure (AZ-900), Node.js 18/20, TypeScript, React + MSAL, Azure AD B2C, Jest, Sentry

Automation & Process Integration — n8n, Webhooks, Selenium, Zabbix, Grafana, Redis/ioredis, OpenAI SDK

Languages — Portuguese (native), English — B1 (EF SET, 2025)